Heirs Insurance Hackathon 2026

Team Information

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1. Executive Summary (1 Page)

Insurance penetration in Nigeria remains critically low despite regulatory efforts and compulsory motor insurance policies. While some fintech platforms such as PalmPay and OPay embed insurance products, adoption remains weak due to lack of trust, poor claims experience, and limited transparency.

A major barrier is not product availability, it is claimed friction and low trust in payout reliability.

InSureAI introduces an AI-powered conversational insurance infrastructure delivered through WhatsApp. The platform enables users to:

- Purchase insurance policies
- Pay premiums through integrated fintech rails
- File claims using voice notes or images
- Receive AI-driven real-time claim evaluation
- Track transparent payout timelines

The core innovation is an Automated Claims Intelligence Engine powered by Artificial Intelligence (AI), Natural Language Processing (NLP), and Computer Vision.

Rather than building another insurance distribution channel, InSureAI focuses on solving the most critical industry problem: trust and claim inefficiency.

The system operates as:

- WhatsApp = Conversational Interface
- Fintech APIs = Payment Rail
- AI Engine = Underwriting & Claims Brain
- Heirs Insurance = Underwriter

By combining conversational access, automated underwriting, and real-time claim triage, InSureAI transforms insurance from a rigid financial product into a seamless daily utility.

This solution addresses:

- Claims Processing
- Underwriting
- Customer Experience
- Product Distribution

and positions Heirs Insurance as a leader in AI-driven insurance transformation.

2. PROBLEM STATEMENT:

Nigeria's insurance penetration rate remains below 3%, despite compulsory policies such as motor insurance.

Key challenges include:

1. Low Trust in Claims Settlement

   Many Nigerians believe insurance companies delay or frustrate claim payouts.

2. Complex Claim Processes

   Manual documentation, long waiting periods, and unclear timelines discourage usage.

3. Poor Adoption

   Insurance is perceived as expensive or unnecessary until an incident occurs.

4. Low Transparency

   Customers lack visibility into how claims are evaluated.

Even where fintech platforms embed insurance, the backend claims process remains largely manual.

This creates a gap:

Distribution has improved.

Trust and claim automation have not.

Without improving claims transparency and speed, insurance penetration will remain low.

There is an urgent need for:

- AI-driven claims triage
- Transparent claim scoring
- Conversational access
- Seamless payment integration

Solving this problem increases adoption, improves retention, and strengthens public confidence in insurance.

3. Proposed Solution

InSureAI is an AI-powered conversational insurance engine deployed through WhatsApp.

3.1. Concept Description

The system allows users to manage insurance activities entirely within a familiar messaging interface.

Core Flow:

1. User interacts with WhatsApp bot
2. AI collects user information for underwriting
3. Risk scoring engine evaluates user profile
4. Coverage recommendation is generated
5. Premium payment processed via fintech API
6. Policy activated instantly
7. Claims processed via AI-driven triage system

The primary innovation lies in the Automated Claims Intelligence Engine.

3.2. Key Features

1. Conversational Underwriting (NLP-Based)

Users answer simple questions via chat or voice notes.

AI extracts:

- Occupation
- Risk indicators
- Coverage intent

The system generates a dynamic risk score and recommends suitable micro-insurance plans.

2. Fintech Payment Integration

Through API integration, users can:

- Pay premiums directly within WhatsApp
- Use wallet providers (e.g., OPay, PalmPay)
- Receive automated renewal notifications

Insurance becomes embedded into daily financial behavior.

3. AI-Powered Claims Triage Engine (Core Innovation)

When a claim is initiated:

Step 1: User sends message or voice note

Step 2: NLP extracts incident details

Step 3: User uploads image evidence

Step 4: Computer Vision evaluates damage

Step 5: Fraud detection model analyzes anomaly patterns

Step 6: Risk classification generated

Claim Outcomes:

- Low Risk → Instant approval & payout initiation
- Medium Risk → Fast-track human review
- High Risk → Investigation queue

Users receive:

- Damage confidence score
- Fraud probability rating
- Estimated payout amount
- Expected settlement timeline

This transparency builds trust and reduces operational burden.

Real-Time Claim Tracking Dashboard (Conversational)

Instead of:

"Your claim is under review."

Users see:

1. Claim status stage
2. Estimated payout time
3. Required next action

3.3. InSureAI differentiates itself by:

- Delivering insurance entirely via conversational interface
- Automating micro-claims through AI triage
- Providing transparent claim scoring
- Integrating payment rails within WhatsApp
- Reducing reliance on manual processing

Most embedded insurance platforms solve distribution.

InSureAI solves trust.

4. Business Impact (Aligned to Criteria)

Revenue Growth

- Increased policy adoption through WhatsApp accessibility
- Higher retention due to faster claims
- Expanded reach into informal sector

Projected impact:

- 20–30% growth in micro-position adoption

Cost Saving

- Reduced manual underwriting costs
- Automated claims triage lowers operational expenses

- Fraud detection minimizes loss ratio

Process & Time Improvement

For Customers:

- Onboarding < 3 minutes
- Claim submission < 5 minutes
- Instant micro-claim approval

For Insurers:

1. Reduced claims backlog
2. Faster risk assessment
3. Improved operational efficiency

5. The user experience is built around familiarity and simplicity.

InSureAI is designed around simplicity, familiarity, and trust. The entire insurance journey takes place within WhatsApp, a platform already used daily by millions of Nigerians. By eliminating the need for a separate application or complex web portal, the system reduces friction and increases adoption.

5.1 Onboarding Experience

The customer journey begins with a simple message:

"Hi"

The AI-powered insurance assistant responds with a guided conversational flow. Instead of presenting lengthy forms, the system asks short, clear questions such as:

- What type of protection are you looking for?
- What do you do for a living?
- What asset would you like to protect?

Users may respond via text or voice note. Natural Language Processing (NLP) converts voice into structured data. The AI risk engine evaluates the user profile in real time and recommends a personalized micro-insurance plan.

Within minutes, the user sees:

- Recommended coverage
- Weekly or daily premium amount
- Coverage benefits
- Simple explanation of what is included

The goal is to complete onboarding in under three minutes.

5.2 Premium Payment Experience

Once the user selects a plan, payment is processed directly within the WhatsApp flow through integrated fintech rails (e.g., wallet providers or bank transfer APIs).

The user receives:

- Instant payment confirmation
- Digital policy summary
- Coverage activation notification

Automated reminders are sent before renewal dates, reducing policy lapses.

This seamless payment integration ensures users never need to leave WhatsApp to manage their insurance.

5.3 Claims Submission Experience (Core Differentiator)

When an incident occurs, the user simply types:

"I had an accident"

or

"My shop was damaged"

The AI assistant responds with guided prompts:

- What happened?
- When did it occur?
- Please send a picture of the damage.

Users may submit text, images, or voice notes. The system processes inputs using:

- NLP to extract incident details
- Computer Vision to assess visible damage
- AI fraud detection to analyze risk patterns

Within minutes, the user receives structured feedback:

- Claim status
- Coverage confirmation
- Estimated payout amount
- Expected settlement time

For low-risk micro-claims, approval and payout initiation occur instantly or within hours.

5.4 Transparency & Trust

Unlike traditional systems where customers receive vague updates such as "Under Review," InSureAI provides structured transparency.

Users can check:

- Current claim stage
- Any missing information
- Estimated processing timeline

This transparency builds trust and reduces anxiety during stressful events.

5.5 Accessibility & Inclusion

The conversational design ensures:

- No complex forms
- No technical jargon
- Voice-first accessibility
- Mobile-only usability
- Support for users with limited digital literacy

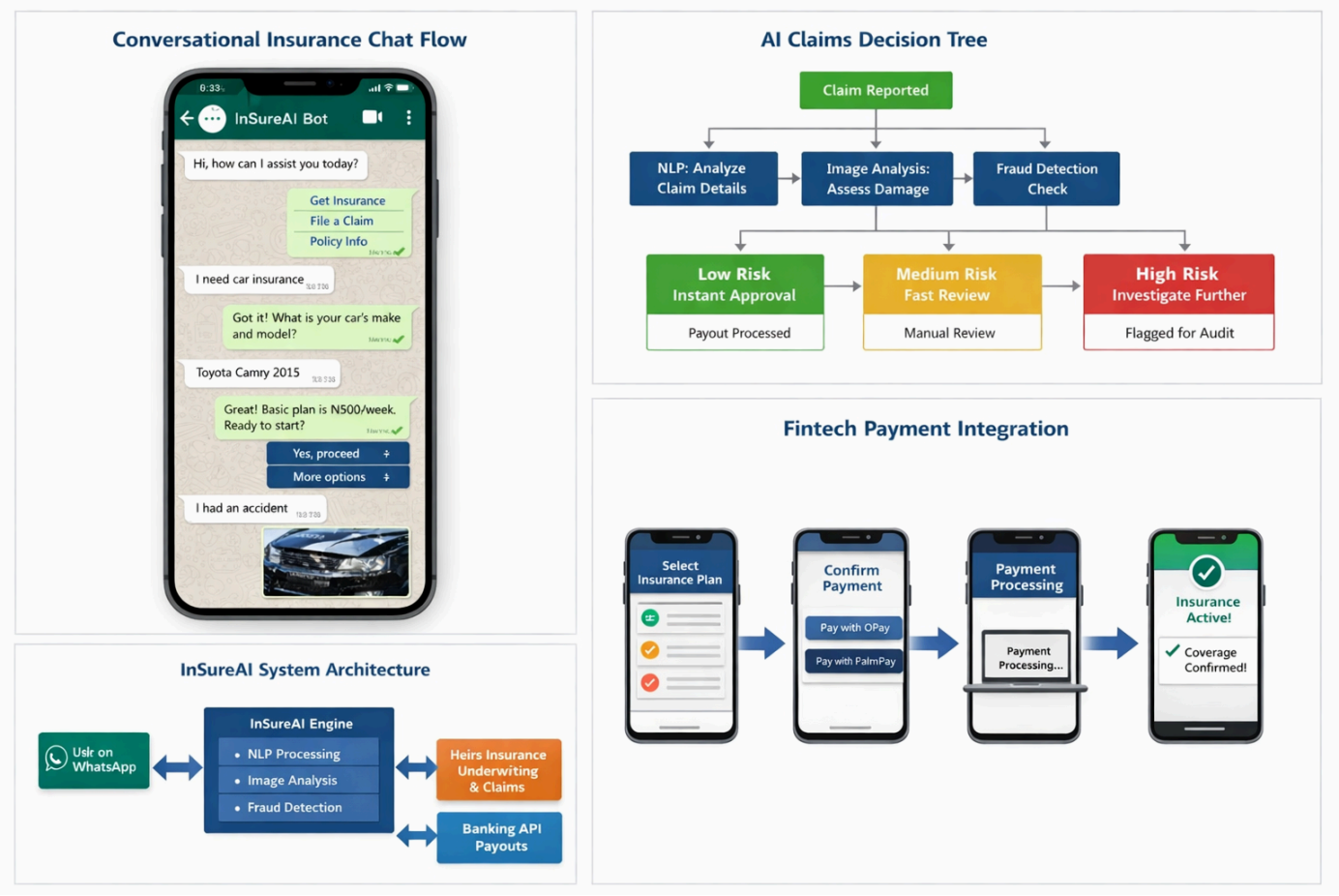
By embedding insurance into a familiar communication channel, InSureAI transforms insurance from an intimidating financial product into a simple, accessible service.

The result is a customer experience that is:

- Fast
- Transparent
- Conversational
- Reliable

- Inclusive

6. VISUAL REPRESENTATION

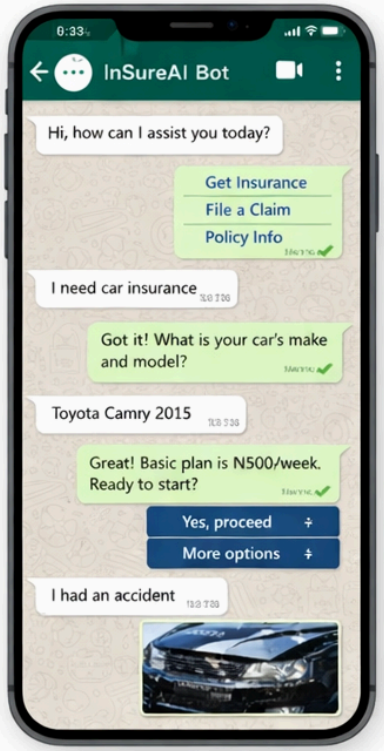


7. CONCLUSION

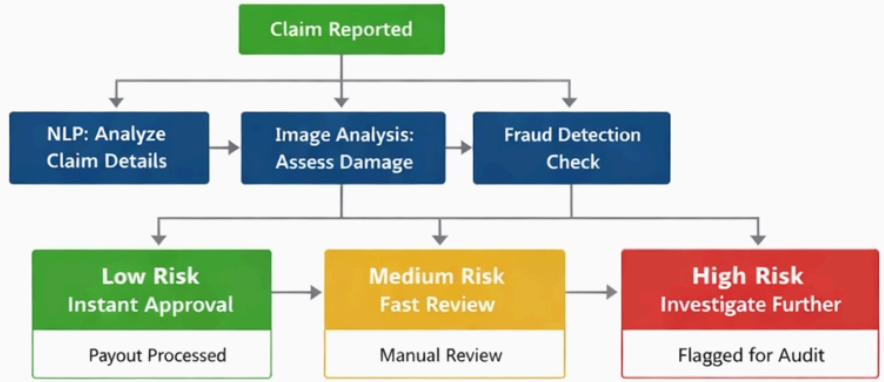
InSureAI transforms insurance delivery by combining artificial intelligence, conversational access, and automated claims processing. By focusing on trust, transparency, and speed, the solution addresses the most significant

barriers to insurance adoption in Nigeria. Through WhatsApp deployment and fintech integration, InSureAI expands accessibility while reducing operational inefficiencies. This solution positions Heirs Insurance at the forefront of AI-driven insurance infrastructure and strengthens its leadership in innovation and customer-centric transformation.

Conversational Insurance Chat Flow



AI Claims Decision Tree



Fintech Payment Integration

InSureAI System Architecture